Day 68 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in History (3): India's 2nd Astronaut & the Mughal 500th Anniversary

READING MATERIAL

Covers a genuine historic milestone — India's second-ever astronaut — and a major historical commemoration: the Mughal Empire's 500th founding anniversary.

Q: Who became the SECOND Indian to travel to space, when, and how?
A: Wing Commander Shubhanshu Shukla, on 25 June 2025 — flying to the International Space Station (ISS) aboard SpaceX's Axiom Mission-4.

Q: Who was the FIRST Indian to travel to space, and when, for context?
A: Rakesh Sharma, in 1984, aboard the Soviet Soyuz T-11 — meaning Shubhanshu Shukla's 2025 flight came 41 years after India's first astronaut.

Q: What major historical anniversary falls in 2026, and what does it mark?
A: The 500th anniversary of the founding of the Mughal Empire — founded on 21 April 1526 (following Babur's victory at the First Battle of Panipat). The empire's architectural legacy includes the Taj Mahal, Agra Fort, and Red Fort.

Q: What was 'Sewa Parv 2025', organised by the Ministry of Culture?
A: A cultural initiative held 17 September - 2 October 2025, featuring art workshops at 75 locations across India.

MOCK TEST (WITH ANSWERS)

1. Who became the second Indian to travel to space, on 25 June 2025?

A. Rakesh Sharma
B. Shubhanshu Shukla ✓
C. Kalpana Chawla
D. Sunita Williams

Explanation: Shubhanshu Shukla. [medium, web-research]

2. Shubhanshu Shukla travelled to the ISS aboard which mission?

A. Gaganyaan-1
B. Axiom Mission-4 ✓
C. Chandrayaan-4
D. Soyuz MS-25

Explanation: Axiom Mission-4 (SpaceX). [medium, web-research]

3. Who was the FIRST Indian to travel to space, in 1984?

A. Rakesh Sharma ✓
B. Shubhanshu Shukla
C. Vikram Sarabhai
D. Homi Bhabha

Explanation: Rakesh Sharma. [easy, web-research]

4. The year 2026 marks the 500th anniversary of the founding of which empire?

A. Maratha Empire
B. Mughal Empire ✓
C. Vijayanagara Empire
D. Chola Empire

Explanation: Mughal Empire (founded 1526). [medium, web-research]

5. The Mughal Empire was founded in 1526, following Babur's victory at:

A. The Battle of Plassey

B. The First Battle of Panipat ✓

C. The Battle of Haldighati

D. The Battle of Talikota

Explanation: The First Battle of Panipat. [medium, web-research]

Part B – Aptitude & Mental Ability: Lowest Common Multiple (LCM)

READING MATERIAL

LCM is 1 of 10 named Aptitude sub-skills. Real exam evidence shows two distinct shapes: algebraic LCM-from-a-clue problems, and 'remainder' word problems — both are taught below as separate methods.

Q: Real exam (2025): If the LCM of 'a' and 'a+2' is 1300, find the LCM of 'a' and 'a+1'.

A: Since a and a+2 differ by 2, their HCF is either 1 or 2. If HCF=2: LCM = a(a+2)/2 = 1300 → a(a+2) = 2600. Solving $a^2+2a-2600=0$ gives a=50 (check: 50×52/2 = 1300 ✓). Now find LCM(50, 51): consecutive integers are always coprime (HCF=1), so LCM = 50×51 = 2550.

Q: Method — 'least number leaving given remainders' (remainders differ from the divisors by a constant): Find the least number which, when divided by 16, 18, and 21, leaves remainders 3, 5, and 8 respectively.

A: Check the gap between each divisor and its remainder: 16−3=13, 18−5=13, 21−8=13 — all the same gap! When that happens, the answer is LCM(16,18,21) – 13. LCM(16,18,21) = 1008. Answer = 1008−13 = 995.

Q: Method — 'least number leaving the SAME remainder in every case': Find the least number which, when divided by 6, 7, 8, 9, and 12, leaves the same remainder 1 in every case.

A: When the remainder is identical across all divisors, the answer is LCM(divisors) + remainder. LCM(6,7,8,9,12) = 504. Answer = 504 + 1 = 505.

Q: Method — 'mixed condition' (a common remainder for some divisors, but exact division for another): Find the least number which, when divided by 5, 6, 7, and 8, leaves remainder 3 in each case, but leaves NO remainder when divided by 9.

A: First find the least number leaving remainder 3 for 5,6,7,8: that's LCM(5,6,7,8)+3 = 840+3 = 843. But we also need it divisible by 9. Keep adding LCM(5,6,7,8)=840 repeatedly to 843 until the result is divisible by 9: 843 (no), 1683 (1683÷9=187, yes!). Answer = 1683.

Q: Method — LCM of numbers given as prime factorizations: Find the LCM of (2×3×5×7) and (3×5×7×11).

A: LCM takes the HIGHEST power of every prime that appears in either number. Primes involved: 2,3,5,7,11 — each appears at most once in either factorization, so LCM = 2×3×5×7×11 (just the union of all prime factors, since none repeat).

MOCK TEST (WITH ANSWERS)

1. If the LCM of 'b' and 'b+2' is 180, and their HCF is 2, find the LCM of 'b' and 'b+1'.

A. 342 ✓
B. 380
C. 361
D. 324

Explanation: LCM = b(b+2)/HCF = 180 → b(b+2) = 360 → $b^2+2b-360=0$ → b=18 (check: 18×20/2=180 ✓). Now LCM(18,19): consecutive integers are coprime, so LCM = 18×19 = 342. [hard, authored]

2. Find the least number which, when divided by 12, 15, and 20, leaves remainders 7, 10, and 15 respectively.

A. 55 ✓
B. 60
C. 65
D. 53

Explanation: Gaps: 12−7=5, 15−10=5, 20−15=5 — all equal, so answer = LCM(12,15,20) – 5 = 60 – 5 = 55. [medium, authored]

3. Find the least number which, when divided by 4, 5, and 6, leaves the same remainder 2 in each case.

A. 58
B. 60
C. 62 ✓
D. 64

Explanation: LCM(4,5,6) = 60. Answer = 60 + 2 = 62. [easy, authored]

4. Find the LCM of (2×2×3×5) and (2×3×3×7).

A. 2×2×3×3×5×7 ✓
B. 2×3×5×7
C. 2×2×3×5×7

D. $4 \times 9 \times 5 \times 7$

Explanation: Take the highest power of each prime: 2^2 (from the first), 3^2 (from the second), 5^1, 7^1. LCM = $2^2 \times 3^2 \times 5 \times 7$, which equals option (A) written out as $2 \times 2 \times 3 \times 3 \times 5 \times 7$. [medium, authored]

5. The LCM of two consecutive integers n and n+1 is always:

A. $n + 1$

B. $n(n+1)$ ✓

C. $n - 1$

D. Cannot be determined

Explanation: Consecutive integers are always coprime (HCF=1), so their LCM is simply their product, $n(n+1)$. [easy, authored]

Part C – General English: Writing Skills: Making Queries, Inferences, Blanks, and Substitutions

READING MATERIAL

Covers drawing reasonable inferences from short scenarios, filling blanks with the correct word/preposition/article, and substituting repeated nouns with appropriate pronouns.

Q: What can be reasonably inferred from: "A person wears a raincoat and carries an umbrella"?

A: It might rain soon — the person's preparation (raincoat, umbrella) reasonably implies they expect rain, even though this isn't explicitly stated.

Q: What can be inferred from: "A person enters a restaurant, looks at the menu, but leaves without ordering"?

A: The person does not want to eat there — the act of leaving without ordering, after reviewing the menu, suggests a decision not to dine there (rather than assuming a specific reason like cost or hunger, which isn't directly supported).

Q: Important nuance: "All birds can fly. Penguins are birds." What can be inferred, and why is this trickier than it looks?

A: The TNPSC-marked answer is 'Penguins CANNOT fly' — which is real-world general knowledge (penguins are flightless), NOT a strict logical deduction from the stated premises (strict logic would actually conclude 'penguins CAN fly', since the premises as stated say ALL birds fly). This question tests recognising that the premise 'All birds can fly' is itself factually false, rather than testing pure formal syllogistic logic — a useful distinction to keep in mind for inference questions.

Q: Fill in the blank correctly: "The train arrived late ____ heavy rain."

A: DUE TO (not 'because' alone, which would need 'because OF'; 'for' and 'although' don't fit the cause-effect relationship here).

Q: Fill in the blank correctly: "I have never seen ____ elephant before" and "She is ____ intelligent girl."

A: AN elephant (vowel sound), AN intelligent girl (vowel sound) — both use 'an' because the following word starts with a vowel SOUND, not because of the letter itself.

Q: Fill in the blank correctly: "Let's go for a walk, ____?" (question tag) and "My brother is interested ____ photography."

A: "Let's go for a walk, SHALL WE?" (the standard question tag for 'Let's' statements). "Interested IN photography" (the fixed preposition that follows 'interested').

Q: Substitution example: "I have two pens. ____ pen is blue and the other is red." What word correctly substitutes for the repeated noun?

A: ONE — "One pen is blue and the other is red" (using 'one... the other' to refer back to the two pens without repeating 'pen' awkwardly).

MOCK TEST (WITH ANSWERS)

1. "A person wears a raincoat and carries an umbrella." What can be inferred?

A. It is sunny

B. It might rain soon ✓

C. The person hates rain

D. The person loves rain

Explanation: It might rain soon. [easy, web-research]

2. "The ground is wet and there are puddles outside." What can you infer?

A. It rained recently ✓

B. Someone washed the ground

C. It will rain tomorrow

D. The ground is always wet

Explanation: It rained recently. [medium, web-research]

3. "A person enters a restaurant, looks at the menu, but leaves without ordering." What can you infer?

A. The food is expensive

B. The person is allergic to food

C. The person does not want to eat there ✓

D. The person was not hungry

Explanation: The person does not want to eat there. [medium, web-research]

4. "All birds can fly. Penguins are birds." What is the TNPSC-marked inference?

A. Penguins can fly

B. Penguins cannot fly ✓

C. Penguins are not birds

D. Penguins swim instead of being birds

Explanation: Penguins cannot fly (based on real-world knowledge overriding the false premise). [hard, web-research]

5. Fill in the blank: "The train arrived late ____ heavy rain."

A. because

B. due to ✓

C. for

D. although

Explanation: due to. [medium, web-research]

6. Fill in the blank: "I have never seen ____ elephant before."

A. a

B. an ✓

C. the

D. some

Explanation: an. [easy, web-research]

7. Fill in the blank: "Let's go for a walk, ____?"

A. isn't it

B. won't we

C. shall we ✓

D. don't we

Explanation: shall we. [medium, web-research]

8. Fill in the blank: "My brother is interested ____ photography."

A. on

B. at

C. in ✓

D. for

Explanation: in. [medium, web-research]

9. "I have two pens. ____ pen is blue and the other is red." Choose the correct substitution.

A. It

B. One ✓

C. Them

D. Those

Explanation: One. [medium, web-research]